Ecosystem Structure, Function, and Dynamics: A Comprehensive Study Guide

This study guide provides a detailed review of ecosystem dynamics, covering structural components, functional processes such as productivity and decomposition, energy flow, ecological succession, and nutrient cycling based on the provided research context.

Ecosystem: Comprehensive Study Notes

1. Introduction to Ecosystems

An **ecosystem** is the functional unit of nature where living organisms interact with each other and their physical environment.

- **Types of Ecosystems:**
    - **Natural:** Terrestrial (forests, grasslands, deserts) and Aquatic (ponds, lakes, rivers, seas).
    - **Artificial / Man-made:** Crop fields, aquariums, zoos, and gardens.
- **Stratification:** The vertical distribution of different species occupying different levels. For example, in a forest ecosystem, trees occupy the top vertical strata, shrubs the middle, and herbs/grasses occupy the bottom layer.

2. Core Functions of an Ecosystem

The components of an ecosystem function as a single unit through four key aspects:

1. Productivity
2. Decomposition
3. Energy Flow
4. Nutrient Cycling

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3. Productivity

Productivity is the rate of biomass (organic matter) production by organisms, expressed in terms of $g^{-2} yr^{-1}$ or $(kcal m^{-2}) yr^{-1}$.

- **Primary Productivity:** The amount of biomass produced per unit area over a specific time by plants during photosynthesis. It is divided into:
    - **Gross Primary Productivity (GPP):** The total rate of organic matter production during photosynthesis.
    - **Net Primary Productivity (NPP):** The remaining biomass after the plant utilizes some energy for respiration (R). NPP is the biomass available for consumption by heterotrophs.
    - **Formula: NPP = GPP - R**.
- **Secondary Productivity:** The rate of formation of new organic matter by consumers.

Note: The annual NPP of the whole biosphere is approximately 170 billion tons, but oceans contribute only 55 billion tons due to limited sunlight and nutrients.

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4. Decomposition

Decomposition is the process of breaking down complex organic matter into simple inorganic substances like CO_2, water, and nutrients.

- **Detritus:** The raw material for decomposition, which includes dead plant remains (leaves, bark) and dead animals, including fecal matter.

Flow Chart of Decomposition Steps: [**Detritus**] ↓ **1. Fragmentation:** Detritivores (e.g., earthworms) break down detritus into smaller particles. ↓ **2. Leaching:** Water-soluble inorganic nutrients seep down into the soil horizon and precipitate as unavailable salts. ↓ **3. Catabolism:** Bacterial and fungal enzymes degrade detritus into simpler inorganic substances. ↓ **4. Humification:** Accumulation of a dark, amorphous substance called humus, which is highly resistant to microbial action and acts as a nutrient reservoir. ↓ **5. Mineralisation:** Microbes further degrade humus to release inorganic nutrients.

- **Factors affecting decomposition:** It is slower if detritus is rich in lignin and chitin, and faster if rich in nitrogen and water-soluble sugars. Warm and moist environments favor rapid decomposition.

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5. Energy Flow

Energy flow in an ecosystem is strictly **unidirectional**; it moves from the sun to producers, and then to various consumers, never reverting back.

- **Photosynthetically Active Radiation (PAR):** Plants capture only **2 - 10%** of PAR to synthesize food.

- **10% Law:** At each trophic level, 90% of energy is degraded as heat, and only **10%** is transferred to the next trophic level. This limits most food chains to 3-4 trophic levels.

Types of Food Chains:

1. **Grazing Food Chain (GFC):** Starts with living plant material (producers) $\rightarrow$ Herbivores $\rightarrow$ Carnivores.
2. **Detritus Food Chain (DFC):** Begins with dead organic matter $\rightarrow$ Decomposers (saprotrophs like fungi and bacteria).

*Note: In terrestrial ecosystems, more energy flows through the DFC, whereas in aquatic ecosystems, the GFC is the major conduit for energy flow. A network of interlinked food chains is called a **Food Web**.*

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6. Ecological Pyramids

Graphical representations of the trophic structure in a food chain (Producers at the base $\rightarrow$ Top consumers at the apex).

- **Pyramid of Number:** Shows the number of individuals. Generally upright, but **inverted** in a tree ecosystem where one tree supports many insects and parasites.
- **Pyramid of Biomass:** Shows the total dry weight. Generally upright in terrestrial habitats, but **inverted** in sea/aquatic ecosystems because the biomass of consumer fishes far exceeds that of the short-lived phytoplankton.
- **Pyramid of Energy: Always upright** and can never be inverted. Energy is always lost as heat at each step when moving to the next trophic level.

Limitations of Ecological Pyramids:

- Does not accommodate food webs (assumes a simple linear food chain).
- Does not account for the same species belonging to multiple trophic levels.
- Saprophytes/decomposers are entirely excluded despite their vital role.

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7. Ecological Succession

The gradual, orderly, and sequential change in the species composition of a given area, eventually leading to a stable **climax community**.

Flow Chart of Succession: [**Pioneer Species** (First to invade)] $\rightarrow$ [**Seral Communities** (Transitional stages)] $\rightarrow$ [**Climax Community** (Stable and mesic)].

- **Primary Succession:** Occurs in completely bare areas where no life existed before (e.g., bare rock, newly cooled lava). It is very slow because soil must first be formed. **Lichens** are typical pioneer species on bare rocks.

- **Secondary Succession:** Starts in areas where organisms previously existed but were destroyed (e.g., burned forests, abandoned farm lands). It occurs much faster because soil and organic matter are already present.

- **Hydrarch Succession:** Starts in water (pioneers: phytoplankton) and proceeds to a mesic (moderate) environment.

- **Xerarch Succession:** Starts in dry, barren areas (pioneers: lichens) and proceeds to a mesic environment.

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8. Nutrient Cycling

The movement of nutrient elements through ecosystem components, also known as biogeochemical cycling. A reservoir is needed to meet deficits caused by imbalances in the influx and efflux rates.

- **Gaseous Cycles:** Reservoir is the **atmosphere** (e.g., Carbon, Nitrogen). *Carbon is added via respiration, burning fossil fuels, and deforestation, and fixed via photosynthesis*.
- **Sedimentary Cycles:** Reservoir is the **Earth's crust** (e.g., Phosphorus, Sulphur). *Phosphorus is released primarily by the weathering of rocks*.

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9. Ecosystem Services

These are the life-support benefits provided by ecosystems, such as purifying air and water, generating fertile soil, cycling nutrients, and providing wildlife habitats.

- Researcher **Robert Constanza** put an average price tag of **$33 trillion a year** on these fundamental services.
- **Soil formation** is the most valuable ecosystem service, accounting for about 50% of this estimated cost.

Section 1: Review Quiz

Instructions: Answer the following questions in 2–3 sentences based on the information provided in the source context.

1. **How is an ecosystem defined in a functional sense, and what are its two primary categories?**
2. **What is stratification in a terrestrial ecosystem, and how is it layered?**
3. **Distinguish between Gross Primary Productivity (GPP) and Net Primary Productivity (NPP).**
4. **Identify the raw materials for decomposition and the primary physical factors that regulate its rate.**
5. **Describe the difference between a Grazing Food Chain (GFC) and a Detritus Food Chain (DFC).**
6. **Why is the pyramid of energy always upright, regardless of the ecosystem type?**
7. **What is a "standing crop," and how is it typically measured?**
8. **Contrast primary succession with secondary succession regarding their starting conditions and speed.**
9. **How do hydrarch and xerarch successions differ in their starting points and ultimate conclusion?**
10. **What are "ecosystem services," and how does their estimated economic value compare to the global Gross National Product (GNP)?**

Section 2: Quiz Answer Key

1. **Ecosystem Definition:** An ecosystem is a functional unit of nature where living organisms interact with one another and their physical environment. It is generally divided into two categories: terrestrial (such as forests or grasslands) and aquatic (such as ponds or estuaries).
2. **Stratification:** Stratification refers to the vertical distribution of different species occupying different levels within an ecosystem. In a forest, trees occupy the top vertical strata, followed by shrubs in the second layer, and herbs and grasses at the bottom.
3. **GPP vs. NPP:** Gross Primary Productivity is the total rate at which organic matter is produced during photosynthesis. Net Primary Productivity is the remaining biomass after accounting for the energy plants use during respiration (GPP – R = NPP), representing the biomass available for heterotrophs.
4. **Decomposition Factors:** The raw material for decomposition is detritus, which includes dead plant remains, animal remains, and fecal matter. The rate of this process is primarily regulated by the chemical composition of the detritus (e.g., lignin and chitin content) and climatic factors like temperature and soil moisture.
5. **GFC vs. DFC:** The Grazing Food Chain (GFC) begins with producers and is the major conduit for energy flow in aquatic systems. The Detritus Food Chain (DFC) begins with dead organic matter and decomposers (saprotrophs), accounting for a larger fraction of energy flow in terrestrial ecosystems.
6. **Upright Energy Pyramid:** The pyramid of energy is always upright because energy is lost as heat at every step as it flows from one trophic level to the next. According to the

10 per cent law, only a small fraction of energy is available to the higher trophic level, making an inverted energy pyramid impossible.

7. **Standing Crop:** A standing crop is the total mass of living material present in a specific trophic level at a particular time. It is measured as either the number of organisms or the biomass per unit area, with dry weight being the more accurate measurement of biomass.
8. **Succession Comparison:** Primary succession begins in lifeless areas where no soil exists, such as bare rock or cooled lava, making it a very slow process. Secondary succession starts in areas where a previous community was destroyed (e.g., abandoned farmland or burned forests), and because soil is already present, it proceeds much faster.
9. **Hydrarch and Xerarch:** Hydrarch succession starts in water or very wet areas, while xerarch succession begins in very dry areas. Despite these different starting points, both processes eventually lead to "mesic" conditions, which are neither too dry nor too wet.
10. **Ecosystem Services Value:** Ecosystem services are the products of ecosystem processes, such as air purification, nutrient cycling, and crop pollination. Researchers estimate these services are worth approximately US $33 trillion annually, which is nearly twice the value of the global GNP (US $18 trillion).

Section 3: Essay Format Questions

The following questions are designed for in-depth analysis and require a synthesis of multiple concepts from the text.

1. **The Mechanics of Decomposition:** Explain the five stages of decomposition (fragmentation, leaching, catabolism, humification, and mineralization) and discuss how environmental conditions can either inhibit or accelerate this cycle.
2. **Energy Flow and Thermodynamics:** Discuss how the first and second laws of thermodynamics apply to the unidirectional flow of energy within an ecosystem, from solar capture by autotrophs to dissipation as heat.
3. **Ecological Pyramids and Their Limitations:** Compare and contrast pyramids of number, biomass, and energy. Include a discussion on why certain pyramids can be inverted and the specific reasons why saprophytes and multi-trophic species are difficult to represent in these models.
4. **Nutrient Cycling Dynamics:** Contrast the gaseous carbon cycle with the sedimentary phosphorus cycle. Identify the primary reservoirs for each and explain how human activities have specifically impacted the global carbon balance.
5. **The Process of Plant Succession:** Describe the progression of a biotic community from pioneer species to a climax community. Detail the specific stages involved in primary succession in a water body, leading to a stable forest.

Section 4: Glossary of Key Terms

Term,Definition
Abiotic,The non-living chemical and physical parts of the environment that affect living organisms and the functioning of ecosystems.

Autotrophs,"Organisms, such as plants and photosynthetic bacteria, that fix solar energy to produce food from simple inorganic materials."
Biogeochemical Cycle,The movement of nutrient elements through the various living (biotic) and non-living (geological/physical) components of an ecosystem.
Climax Community,A community that has reached a near-equilibrium state with its environment through the process of ecological succession.
Detritivores,"Organisms, such as earthworms, that break down detritus into smaller particles during the process of fragmentation."
Ecosystem,A functional unit of nature where living organisms interact among themselves and with their surrounding physical environment.
Humification,"A step in decomposition that leads to the accumulation of humus, a dark-colored amorphous substance resistant to microbial action."
Leaching,The process by which water-soluble inorganic nutrients move down into the soil horizon and precipitate as unavailable salts.
Mineralization,The process by which some microbes further degrade humus to release inorganic nutrients.
Pioneer Species,"The first species to invade a bare area, such as lichens on bare rock or phytoplankton in water."
Saprotrophs,"Decomposers, mainly fungi and bacteria, that meet their energy needs by degrading dead organic matter using digestive enzymes."
Sere,The entire sequence of communities that successively change in a given area during ecological succession.
Standing State,"The total amount of nutrients, such as carbon, nitrogen, and phosphorus, present in the soil of an ecosystem at any given time."
Trophic Level,A specific place or functional level occupied by an organism in a food chain based on its source of nutrition.